

Splunk: Detecting Nmap Port Scanning

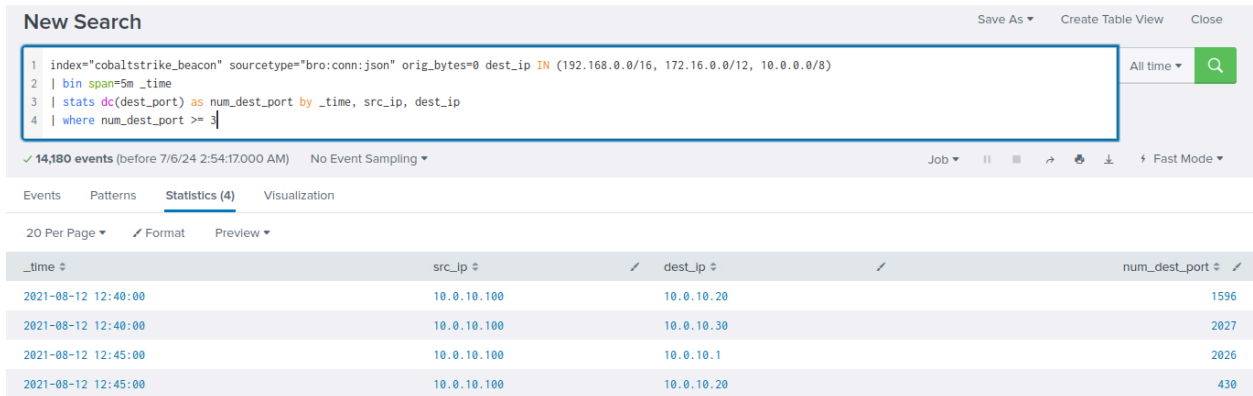
Objective

The objective of this lab is to detect Nmap port scanning activities using Splunk.

Steps

1. Nmap Port Scanning Detected

Utilized Splunk to identify and detect instances of Nmap port scanning activities within the network logs.



The screenshot shows the Splunk search interface. The search bar contains the following query:

```
1 index="cobaltstrike-beacon" sourcetype="bro:conn:json" orig_bytes=0 dest_ip IN (192.168.0.0/16, 172.16.0.0/12, 10.0.0.0/8)
2 | bin span=5m _time
3 | stats dc(dest_port) as num_dest_port by _time, src_ip, dest_ip
4 | where num_dest_port >= 3
```

The search results show 14,180 events. The table below displays the results for the search.

_time	src_ip	dest_ip	num_dest_port
2021-08-12 12:40:00	10.0.10.100	10.0.10.20	1596
2021-08-12 12:40:00	10.0.10.100	10.0.10.30	2027
2021-08-12 12:45:00	10.0.10.100	10.0.10.1	2026
2021-08-12 12:45:00	10.0.10.100	10.0.10.20	430

Conclusion

Detecting Nmap port scanning is crucial for network security monitoring. By leveraging Splunk's capabilities to analyze network logs, organizations can promptly detect and respond to potential reconnaissance activities, enhancing their ability to protect against unauthorized access and potential security breaches.